

SMART INDIA HACKATHON 2026 · AGRICULTURE, FOODTECH & RURAL DEVELOPMENT

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See deeper. Save grain.

Deep-stack grain monitoring and early-warning system — **detect · predict · protect**

Spoilage starts where inspection stops.

1 · LOCALISED MOISTURE

A small wet pocket forms

Roof leaks, wet bags, condensation or a damp corner raise moisture in one small region inside a stack — not across the whole of it.

2 · HIDDEN DETERIORATION

Conditions turn favourable

Higher water activity allows fungal growth. Dry matter is lost and, with **Aspergillus flavus**, aflatoxin risk can develop **before any mould is visible**.

3 · DAMAGE SPREADS SILENTLY

Nobody can see it happen

Conventional inspection samples what it can reach. Deterioration begins where it cannot.

THE REAL CHALLENGE

Detecting hidden deterioration deep inside grain stacks

3.89–5.92%

Indian post-harvest cereal loss
NABCONS, reported to Parliament 2024

Spoilage breathes.

Gas arrives first.

One reaction, three products

Grain is alive, and so are the fungi on it. Both consume sugar and oxygen and release **CO₂, water and heat — from a single reaction.**

But those three products travel very differently. CO₂ escapes through the air gaps between kernels. Heat must conduct through the kernels themselves, and grain is an excellent insulator.

A compost heap, buried

You smell a compost heap across the garden long before you could feel its warmth by touching it. A spoiling pocket is the same event — smaller, slower, and out of sight.

TIME TO TRAVEL THE SAME 0.4 M

CO ₂	0.9 DAYS
Heat	15.4 DAYS
Moisture	123.5 DAYS

Computed as **t = L²/D** from published transport coefficients. Because both times scale with distance squared, **the ratio holds at any separation.**

16.7×
gas leads heat — about two weeks of warning

We are not the first to see this.

We are the first to reach it.

DIRECT PRECEDENT

Detection of a Developing Hot Spot in Stored Corn with a CO₂ Sensor

Ileleji, Maier, Bhat & Woloshuk · *Applied Engineering in Agriculture* 22(2):275–289, 2006 · ASABE

*Induced hot spot in a 12.5 t bin. Headspace CO₂ rose from 500 to 1,500–2,300 ppm across three trials. CO₂ sensors “**detected spoilage earlier than temperature cables**” — and “temperature cables alone might not be a reliable indicator.”*

ESTABLISHES THE FIELD

Integrated approach on stored grain quality management with CO₂ monitoring

Ramachandran · *J. Stored Prod. Res.* 96:101950, 2022 · 122 refs

VALIDATES THE SENSING TRIAD

Real-time CO₂ monitoring for early detection of spoilage and mycotoxin contamination

Kiaitsi, Zanardi, Sulyok, García-Cela et al. · *J. Sci. Food Agric.*, 2025

BACKS THE GROWTH MODEL

Modelling temperature and water activity on *A. flavus* growth and aflatoxin production

Norlia, Jinap et al. · *Int. J. Food Microbiol.* 335:108836, 2020

From “something smells” to “L2 at 2 m. Act in 6 days.”

01 SENSE Interstitial CO₂, humidity and temperature at nine ports across three depths on three lances.	02 INTERPRET Humidity becomes water activity; water activity plus temperature becomes a growth forecast.	03 DECIDE Composite risk score, ranked ports, and an action card carrying a deadline.	04 VERIFY Post-treatment signature decay confirms the intervention actually worked.
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The output is not an alarm

It is **a port, a reason, and a deadline**. A reading of 4,200 ppm tells a godown manager nothing they can act on.

The loop closes — and nobody else’s does

Current practice verifies nothing. A failed fumigation and a successful one look identical until infestation returns weeks later.

Moisture is the total.

Water activity is the spendable part.

MOISTURE (WB)	TEMP	A _w	MEANING
12 %	30 °C	0.657	Safe — growth not supported
14 %	25 °C	0.774	Safe
16 %	30 °C	0.875	Active growth
18 %	25 °C	0.926	Rapid
20 %	30 °C	0.963	Severe

The core idea

Not all water in a kernel is usable — some is chemically bound to starch and protein, and fungi cannot reach it. **Water activity is the free fraction, 0 to 1.** At equilibrium it *is* the humidity of the air between the kernels — which is why an ordinary humidity sensor measures the governing variable.

0.82 — THE GROWTH FLOOR

The **conservative end of the published range** for *A. flavus*. Reported minima vary with strain and substrate; we deliberately took the lowest credible value, because a warning system should alert early rather than late.

0.85 — THE TOXIN LINE

Growth begins at 0.82, but **aflatoxin production needs conditions wetter still.** Two thresholds, not one.

TIME TO VISIBLE MOULD AT 30 °C

a _w 0.95	6.7 DAYS
a _w 0.90	10.8 DAYS
a _w 0.87	17.4 DAYS
a _w 0.85	28.9 DAYS
a _w 0.83	86.8 DAYS

A pocket only has to cross about **15 % moisture**. Then the clock starts — and the clock is what we report.

You can stop the fungus. You cannot unmake the toxin.

*Aflatoxin is a liver carcinogen. Drying does not destroy it. Cooking does not destroy it — it is heat stable. **Once aflatoxin is in a consignment, that grain's fate is decided.***

DIAGNOSIS IS TOO LATE TO ACT ON

A system that tells you grain is already contaminated has told you nothing you can do anything about.

SO THE PROBLEM IS PREDICTIVE

The only intervention that changes the outcome happens **before** the fungus grows. We forecast the conditions rather than confirm the damage.

AND WE STATE THE LIMIT

We do not measure toxin. We predict the conditions that create it, and we say so in every output the system produces. Confirming toxin needs a laboratory assay.

The problem isn't diligence. It's reach.

- 1

Only reaches the outside

A spear reaches roughly **0.4 m**. A godown stack is metres deep. The middle is not reachable.
- 2

Problems cluster

Insects and moisture gather in pockets. A clean sample here says nothing about a pocket two metres away.
- 3

Early stages are invisible

Eggs and larvae develop **inside** kernels. A sample with no adults can still be heavily infested.

Why the chemistry is failing too

Indian *Tribolium castaneum* shows **2.11 to 70.94-fold** phosphine resistance (*Sci. Reports* 13:16497, 2023). Blanket dosing selects for exactly this — and fumigation kills insects; it does not un-rot grain, and it cannot remove aflatoxin.

*“A spear reaches forty centimetres. Our hotspot is at **one and a half metres** from the nearest reachable face. No amount of diligence gets a spear to that location.”*

This is not a criticism of inspection. It is precisely the reason an interior probe needs to exist.

We weakened our own headline number on purpose.

ASSUMPTION	OUR VALUE	WHY IT IS GENEROUS TO THE BASELINE
Inspection interval	every 14 days	Fortnightly is the generous end — many facilities are monthly or slower
Spear reach	0.4 m	The physical limit of a bag spear
Finding a pocket in reach	60 %	Problems cluster; six in ten is reasonable, not harsh
Surface mould	always found	We assume inspectors never miss visible mould — not once
Musty smell / warm bags	also found	A second detection route we added to help the baseline

Our first model had conventional inspection **never** finding this pocket — arguably what the guidance implies. We thought that looked rigged, so we added smell-and-warmth detection. **That is where day 42 comes from.** Without it our gap would look considerably bigger.

SAME EVENT. TWO TIMELINES.

CONVENTIONAL

day 42

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day 0.8

Simulation result on an identical injected event. Actual mould onset: day 9.7.

Four layers. One answer: where, how bad, how long.

SENSE

9 ports · 3 lances × 3 depths

One shared NDIR analyser across all nine sampling points. SHT31 humidity, because relative humidity at equilibrium **is** water activity.

INTERPRET

Isotherm → growth kinetics

Humidity becomes water activity; water activity and temperature feed a cardinal-value growth model giving **days-to-onset per zone**.

NETWORK

One node per stack · offline-first

Zero external dependencies. Runs entirely with WiFi off, because rural godown connectivity **cannot be assumed**.

SEE

Ranked ports · risk map

Every hotspot with a coordinate, a water activity, a deadline and an accumulated dry-matter loss.

The gotcha we designed around

Every NDIR sensor **auto-rezeros to the lowest value it sees**. Inside grain it never meets fresh air — so ABC must be disabled in firmware, or the sensor quietly erases the very signal we exist to detect.

Capacity is outpacing the ability to monitor it

Under the World’s Largest Grain Storage Plan, **500+ PACS godowns** are under construction. None are instrumented today.

We chose the explainable detector.

✗ **Not a fixed threshold**

Godown temperature swings seasonally. Any absolute limit either cries wolf in summer or sleeps through winter.

✗ **Not all nine ports together**

Depth stratification is real and about **1,000 ppm** — deep grain is furthest from a venting face. Comparing all nine buried the signal completely.

✓ **Depth-matched differential**

Each port is compared **only against the other lances at its own depth**, on the same analyser. This cancels sensor drift and slow ventilation changes at the same time — and it is the reason a real pocket stands out at all.

THE MEASURED SEPARATION

CLEAN GRAIN PEAKS AT

1.5 σ

WE ALERT AT

3.0 σ

A REAL HOTSPOT REACHES

4.2 σ

Sustained over three consecutive readings — **one noisy sample is never an alarm.**

State the boundary before they find it.

CLAIM	STATUS
Two sealed maize jars — one at storage moisture, one deliberately wetted	YES
SHT3x reading headspace on the same ESP32 firmware built for the field probe	YES
Reading fed into the same decision-engine code as the simulation	YES
Dry jar below the 0.82 growth floor; wet jar above it, with a predicted onset	YES
“We watched grain spoil” — sealed 15 hours, fungus needs days	NO
“We measured CO ₂ ” — sensor not sourced yet	NO

Being precise about that boundary is stronger than blurring it — and it is the difference between a team that gets **believed** and one that gets **caught**.

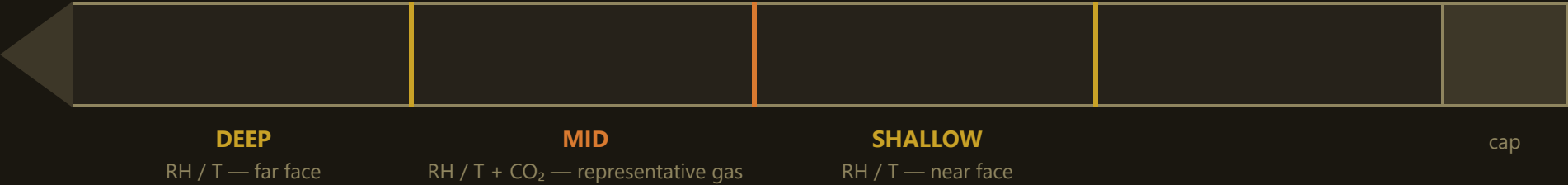
THE WORKING DIGITAL TWIN

Not a mockup. You can break it yourself.

- Explicit finite-difference solver — CO₂, heat and moisture through a 4.0 × 2.5 m stack at 10 cm cells
- Modified Chung–Pfoest isotherm (ASAE D245) and a cardinal-value predictive mycology model
- Virtual sensors carrying datasheet-accurate noise, so the detector is proven on imperfect data
- A conventional-inspection arm running in parallel on the same injected event
- **Self-checking** — headless assertions must pass before every session

Three lances. Nine eyes.

PROBE LANCE — SENSOR STATIONS AND PLACEMENT RATIONALE



Schematic — radial dimension exaggerated for legibility. Mid-depth carries the CO₂ port because it avoids edge dilution and gives the most representative interstitial reading.

Shared analyser

Ports feed a manifold into **a single NDIR instrument**. The expensive part serves nine sampling points — and every depth comparison is made by the same instrument against the same baseline, so drift cancels.

Non-destructive

Perforated lance driven between bag courses. **Removable and reusable** across stacks — which matters because stacks are built and dismantled.

HARDWARE, COSTED

SCD30 NDIR CO ₂	₹5,000
3× SHT3x	₹750
ESP32 ×2	₹600
TCA9548A multiplexer	₹250
Supply, cable, connectors	₹1,050
Total	₹7,650

Inside the ₹19,800 proposal budget.

All 3.3 V I²C by design

ESP32 GPIO is **not 5 V tolerant**, so every part was chosen at 3.3 V — no level shifters, nothing destroyed by a wiring slip. A TCA9548A multiplexer resolves the SHT31 address collision that would otherwise stop three sensors sharing one bus.

Detect early enough and you never need the chemical.

TIER 1 · AUTOMATIC

Aerate

Drive existing godown aeration over the flagged zone. Reversible, non-toxic, cheap. The system watches CO₂ and water activity fall and **confirms it worked**.

TIER 2 · HUMAN HOURS

Turn / isolate

Pull the affected bags, dry and separate them, return the sound grain. **Only possible because you know which bags.**

TIER 3 · LICENSED OPERATOR

Targeted fumigation

Fumigate **that stack, not the shed**. Less chemical, less exposure, less selection pressure for resistance.

THEN · AND NOBODY ELSE DOES

Verify

If the CO₂ anomaly does not fall, the system **escalates rather than assuming success**. Current practice never checks at all.

Late detection leaves only Tier 3

Applied to everything, on a chemical that resistance is already eroding.

Early detection resolves most events at Tier 1

The fumigant becomes the exception rather than the routine.

The sensing hardware is commodity. The intelligence is the hard part.

16.7×

gas leads heat — the physics we build on, and it is published

₹7,650

prototype hardware, inside a ₹19,800 budget

500+

PACS godowns under construction, none instrumented today

NEXT

Source the CO₂ analyser

Run the two-column lead-time experiment that converts our modelled figure into a measured one.

THEN

Coverage analysis

Determine what percentage of hotspots three lances at this spacing detect, and how quickly — the number that justifies the geometry.

THEN

Single-godown pilot

A documented lead-time and chemical-use comparison against that facility's existing practice.